

Pursuing *minor degree in Artificial Intelligence and Machine Learning* from C-MInDS, IITB

SCHOLASTIC ACHIEVEMENTS

- Achieved an impressive **99.78** percentile in JEE Mains, ranking among the top **0.2%** among **1.4M+** candidates ['24]
- Achieved an **All India Rank 1432** in JEE Advance among more than **180,000** highly competitive candidates ['24]
- Qualified **IOQM** (Indian Olympiad Qualifier in Mathematics) conducted by **MTA** (Maths Teacher Association) ['24]
- Qualified **InCho, InPho Stage-1** with Centre Top **10%** conducted by Homi Bhabha Centre Science Education ['24]
- Secured **1st position** in State Camp of **Vigyan Vidyarthi Manthan** exam conducted by NCERT and selected for 3-day **National Camp** among only **28** students from India, held at **Indian Institute of Technology BHU** ['23]
- Got International **Rank 15** in **ISTSE** (Indian School Talent Search Examination) conducted by Edu Excellentia ['22]
- Awarded with **Special Achievement Certificate** for multiple Olympiads by **Silverzone Olympiads** ['21, '22, '23]

TECHNICAL EXPERIENCE

Junior Engineer | *SeDriCa, UMIC IITB* | Prof. Archak Mittal

(Oct '24 - Present)

SeDriCa is a specialized team of 40+ students developing **India's 1st Self Driving Car** with **Level 5** autonomy

- Serving as Junior Engineer in **Motion Planning & Simulations** responsible for validating path finding algorithms
- Acquired proficiency in **Robot Operating System (ROS2)**, **GitHub** and **Docker** through specialized training
- Simulated and benchmarked algorithm performance on **Autodrive** and **F1TENTH Gym ROS** environments

KEY PROJECTS

Summer Training in Advanced Robotics | *SeDriCa* | Unmesh Mashruwala Innovation Cell (May '25 - June '25)

Collaborated in a team of 7 to develop an autonomous car that detects and shoots targets while avoiding obstacles

- Engineered a **completely autonomous** 3D printed differential drive robot, with an elastic energy powered cannon
- Implemented **A* algorithm** for obstacle-aware path planning while exploring algorithms like **RRT, BFS and DFS**
- Built **ROS 2** motion planning node on **Raspberry Pi 5** able to handle **20-24 FPS** data from **NVIDIA Jetson**
- Implemented **PID control** in C++ through ROS reducing cross-track error by **65%** and peak overshoot by **70%**

Conquering Competitive Programming | *Summer of Code* | Web and Coding Club, IIT Bombay (May '25 - Jul '25)

- Consistently practiced and mastered key algorithms and data structures like **Dynamic Programming and Graphs**
- Improved **problem-solving speed and accuracy** by completing rigorous assignments and timed practice sessions
- Solved **100+** problems on platforms like **Leetcode** and **Codeforces** using algorithms from **Number Theory**, **Graph Theory**, and **Geometry** with optimized solutions and detailed **Time and Space complexity analysis**

Decoding The Basket | *Stratify* | Analytics Club

(Aug '25)

- Achieved a **0.75 silhouette score** using **K-means clustering** model to get consumer buying behaviour patterns
- Formulated logic for a **product recommendation engine** using association rule mining to identify bundles with high lift (> 0.1) and confidence (> 0.5), enabling targeted cross-selling strategies to increase customer basket size
- Interpreted complex visualizations using **Matplotlib** and **Seaborn**, including a **heatmap** of order distribution by hour and day, and a **network graph** to visually represent the relationships between products purchased in a basket

Smart India Hackathon | Ministry of Education

(Aug '25 - Present)

Problem Statement : Accelerating High-Fidelity Road Network Modeling For Indian Traffic Simulations

- Significantly improved **pre-deployment** testing by simulating complex and hazardous road scenarios like potholes
- Currently making an **LLM-powered prompt engine** that converts text inputs into realistic traffic environments
- Manual Effort Automated: Behaviours are generated in seconds via **Reinforcement Learning** and **Game Theory**
- Engineered to work natively with RL Toolbox and **Automated Driving Toolbox** for scalable system validation

Guide: Prof. Joseph John Prof. PC Pandey | MS101: Makerspace

- Built a controller to successfully demonstrate flight of a **quadcopter** using **Arduino Nano** and **ESP32-WROOM**
- Designed and fabricated a **custom joystick body** in **Fusion 360**, ensuring ergonomic usability for drone operation
- Developed and tested the **joystick PCB** in **EasyEDA**, integrating sensors and circuits for accurate signal capture
- Calibrated the joystick and implemented wireless control via **ESP32**, enabling precise real-time drone navigation

Heart Disease Prediction | Self Project , Machine Learning

(Sept '25)

- Built a Custom classifier for Real-time prediction of Cardiac Disease Present vs Absent, achieving over **80%** reliability
- Utilized the UCI Heart Disease dataset to build a **binary classification model** for cardiovascular risk prediction
- Achieved **85.1%** training and **81.9%** test accuracy using **Logistic Regression** after hyperparameter tuning
- Utilized **Scikit-learn**, **Pandas**, and **NumPy** to streamline feature handling and model training to boost efficiency

Marketing Spend Optimization & Trend Analysis | Course Project

(Jan'25 - May'25)

Guide: Prof. Mayank Parekh | SOM101: Introduction to Management

- Analyzed **8+** Netflix marketing strategies via **SWOT** analysis, including AI personalization and guerrilla campaigns
- Devised a year-wise **budget model** across **7** categories, identifying trends like the **2020 dip** and **2023–24 growth**
- Identified initiatives in regional localization and influencer marketing, projecting a **15 to 20% rise in engagement**

POSITION OF RESPONSIBILITY

Mentor in Amazon Future Engineer Challenge | Solve for Society

(Aug '25 - Sep '25)

Program started by Amazon and Innovation Story empowering students to develop tech solutions for real-world problems

- Among **15** students from IIT Bombay to guide 2 teams to solve real-world challenges providing technology solutions
- Designed **20-day structured technical pipeline** to prototype innovations and present outcomes to a judge panel

TECHNICAL SKILLS

- **Programming:** C, C++, HTML, LaTeX, Python, CSS
- **Software:** ROS2, Fusion 360, GitHub, Microsoft Office, Arduino, Gazebo Classic
- **Libraries and Framework:** SciPy, NumPy, Pandas, PyTorch, TensorFlow, Matplotlib, Keras, OpenCV

KEY COURSES UNDERTAKEN

Data Science and Machine Learning	Computer Programming and Utilization – CS101, Programming for Data Science * - DS203, Numerical Analysis – ME225
Mathematics & Physics	Integral & Differential Calculus – MA105, Linear Algebra and Differential Equations – MA110
Others	Introduction to Management – SOM101, Introduction to Economics* – EC101, Makerspace – MS101

EXTRA-CURRICULAR ACTIVITIES

* Courses to be completed by Dec 2025

Sports	• Represented my district in Chess at State level in the 55th Punjab State games and qualified for the prestigious Open Nationals by ranking 10th amongst 200+ players • Acheived 1st position in Department Chess Open organised by Mechanical Association • Three-Time Position holder in 13th,15th and 16th Punjab Kesri Chess Championships • Won Silver medal in 'Freshiesia' Kho Kho tournament for Hostel 16 in first year • Selected in 1 year long Chess Coaching under IM Sharad Tilak at NSO in IIT Bombay
Speaking	• Represented IIT Bombay in Asian Parliamentary Debating at Symbiosis , Pune • Clinched Gold in Extempore and Silver in Declamation in inter - schol competition
Miscellaneous	• Mentored a team of 4 freshmen in XLR8 by Electronics and Robotics Club , IIT Bombay • Completed 4-week Learners' Space workshop in Finance 101 by Finance club,IIT Bombay • Awarded Silver medal in SOF National Science Olympiad for excellent performance